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Efficient Sampling and Reconstruction of Traveltime Volumes for Kirchhoff Depth Migration

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Abstract

This study introduces a method for variable grid generation aimed at improving the efficiency of traveltime calculations in seismic imaging. The approach dynamically adjusts grid density based on interpolation error metrics, significantly reducing computational and storage demands while maintaining high precision. Numerical tests demonstrate the effectiveness of this method in optimizing grid configurations, thereby enhancing the overall performance of seismic imaging processes. The method's adaptability to varying subsurface conditions and its ability to maintain high accuracy make it a valuable tool for geophysical applications.

Introduction

Seismic imaging is a fundamental technique in geophysics used to visualize subsurface structures by analyzing the reflections of seismic waves [1]. These waves are generated by controlled sources, such as dynamite explosions or vibrator trucks on land, and air guns in marine environments [2]. The reflected waves are recorded by receivers, which can be geophones on land or hydrophones in marine settings. The data collected from these reflections are processed to create detailed images of the subsurface, revealing geological features such as faults, folds, and stratigraphic layers.

One of the primary techniques in seismic imaging is seismic tomography, which involves the use of seismic waves to create a three-dimensional model of the subsurface. This method is particularly useful for identifying variations in seismic velocity, which can indicate different rock types and fluid contents. Seismic tomography is widely used in oil and gas exploration to locate hydrocarbon reservoirs and in earthquake seismology to study the Earth's interior [3]. Another important technique is seismic migration, which is used to correct the positions of reflectors in seismic images. Migration algorithms reposition the recorded seismic events to their correct locations in the subsurface, improving the accuracy of the final image [4]. Kirchhoff depth migration, a specific type of migration, is particularly effective in areas with complex geology, as it accounts for variations in seismic velocity and can handle steeply dipping reflectors [5].

Accurate traveltimes calculations are crucial for both seismic imaging techniques to construct precise subsurface images, which are essential for various applications, including oil and gas exploration, reservoir characterization, and geotechnical investigations. Traveltimes calculations involve determining the time it takes for seismic waves to travel from the source to the receiver, passing through various subsurface layers [6]. These calculations are fundamental to seismic imaging as they directly influence the accuracy of the final subsurface image. Traditional methods for traveltimes calculations involve creating traveltimes tables for a uniform surface grid. This approach can be computationally intensive and storage-demanding, especially in areas with significant subsurface heterogeneity and complex topography [7]. The uniform grid method requires a dense grid to ensure interpolation precision, leading to high computational and storage costs. In complex geological settings, the need for a fine grid becomes even more pronounced, further exacerbating these challenges.

Grid generation is a critical aspect of traveltimes calculations. The grid defines the points at which traveltimes data are calculated and interpolated. A well-designed grid can significantly enhance the efficiency and accuracy of traveltimes calculations. However, traditional uniform grids often fail to optimize computational resources, as they apply the same grid density across the entire area, regardless of the complexity of the subsurface. To address these challenges, this study introduces a novel variable grid method that dynamically adjusts grid density based on interpolation error metrics. The proposed method refines the grid in regions where interpolation errors exceed a predefined threshold, ensuring accuracy without the need for excessive grid density in areas with minimal interpolation errors. This approach optimizes computational and storage resources, making traveltimes calculations more efficient and accurate.

Methods

Kirchhoff Depth Migration

Kirchhoff depth migration (KDMIG) is a widely used seismic imaging technique that involves the repositioning of recorded seismic events to their correct locations in the subsurface. This method is particularly effective in areas with complex geology, as it accounts for variations in seismic velocity and can handle steeply dipping reflectors.

The general formula of KDMIG can be given by the following integral expression [5]:

$$I(\xi) = \int W(\xi, g, s) D[t = t_D(\xi, g, s), g, s] dg ds, \quad (1)$$

where $I(\xi)$, defined in a three-dimensional space $\xi = (z_\xi, x_\xi, y_\xi)$, is equal to the integral of the data values $D(t, r, s)$ evaluated at the time $t_D(\xi, g, s)$ and weighted by an appropriate factor $W(\xi, g, s)$. The amplitude factor should also include a one-dimensional convolution by a derivative filter, with the order of the derivative depending on the dimensionality of the integration.

In Kirchhoff depth migration, the time delays $t_D(\xi, g, s)$ are typically computed numerically using ray tracing or an Eikonal equation solver. Figure 1 illustrates how the downgoing and the upgoing rays are traced through an interval velocity function $v(z, x, y)$. The resulting time delays can be expressed as:

$$t_D(\xi, g, s) = t_s[\xi, s, v(z, x, y)] + t_g[\xi, g, v(z, x, y)] \quad (2)$$

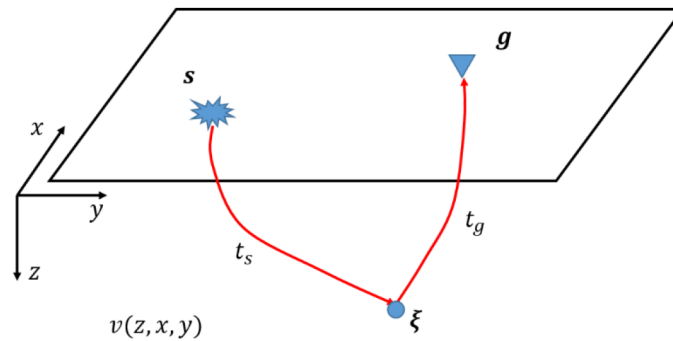


Figure 1—Graphical illustration of the ray tracing necessary to compute the time delays used for KDMIG of a single trace.

where t_s is the time delay from the source location s to the image point ξ , and t_g is the time delay from the same image point ξ to the receiver location g . For a fixed source or a fixed receiver with varied image points, t_s and t_g can be regarded as two traveltimes tables or traveltimes volumes.

There are various implementations of Kirchhoff depth migration, but they generally share the following major steps [8]:

1. Computing the traveltimes tables from the source and from the receiver to the image points
2. Preprocessing the input seismic traces
3. Migrating the seismic traces according to the traveltimes information
4. Postprocessing and outputting the migration results

Considering that computing the traveltimes table on-the-fly during migration of each trace is computationally expensive, the traveltimes tables are precomputed at coarsely sampled uniform surface and subsurface grids and stored on disk. For each input trace, KDMIG performs the following operations:

1. Reading two traveltimes tables from the disk, interpolate these two traveltimes tables to the source or the receiver location.
2. Interpolating the traveltimes tables from a coarse subsurface grid to a fine grid.
3. Spraying each sample of the input trace to the image volume according to the traveltimes information.

Since traveltimes interpolation is relatively efficient compared to ray tracing, the computational cost and disk consumption depend on the number of traveltimes tables to be computed and stored. Traditional methods for traveltimes calculations involve creating traveltimes tables for a uniform surface grid. This approach can be computationally intensive and storage-demanding, especially in areas with significant subsurface heterogeneity and complex topography. The uniform grid method requires a dense grid to ensure interpolation precision, leading to high computational and storage costs. In complex geological settings, the need for a fine grid becomes even more pronounced, further exacerbating these challenges.

To address these issues, the proposed variable grid method dynamically adjusts grid density based on interpolation error metrics. This ensures accuracy without the need for excessive grid density in areas with minimal interpolation errors, optimizing computational and storage resources and making traveltimes calculations more efficient and accurate.

Variable Grid Generation

The proposed method begins with a coarse grid structure and selectively refines grid cells where interpolation errors surpass a predetermined threshold. The error evaluation at a cell's midpoint involves statistical analysis of discrepancies between ray-traced traveltimes data and interpolated values. This discrepancy can be quantified through various statistical measures, such as mean, median, or specific

percentiles of the difference, either directly from the traveltimes tables or transformed into slowness or celerity domains.

The slowness can be calculated using the equation:

$$s = t/l, \quad (3)$$

where s is the slowness, t is the traveltimes, l is the straight-line distance from the surface grid point to each of the subsurface grid points. The celerity can be calculated by the following equation [9]:

$$c = l/t, \quad (4)$$

where c is the celerity.

Upon identifying a grid cell with an interpolation error exceeding the threshold, the method subdivides the cell, introducing additional grid points to increase local grid density. This iterative refinement process ensures that interpolation errors are minimized across the entire grid. The algorithm continually assesses the interpolation error against the error bound parameter, refining the grid until the errors are within acceptable limits.

The error bound parameter can be determined using the equation:

$$\Delta t = \frac{1}{2f_{max}}, \quad (5)$$

where f_{max} is the seismic data's maximum frequency. f_{max} is usually below 100 Hz in seismic exploration. Therefore, the interpolation error should be less than 5 milliseconds. If the error is evaluated in the slowness domain, we can approximate the maximal distance as 10 km, then the slowness error should be less than 5×10^{-4} s/km.

Variable Grid Generation Procedure

Below delineates the comprehensive procedural sequence for the error-bounded variable grid generation method:

1. Central Point Interpolation: For each quartet of adjacent grid points forming a quadrilateral, estimate the central value via interpolation using the values at the vertices.
2. Interpolation Error Assessment: Compare the interpolated central value against the precise value derived from ray tracing. If the error exceeds a designated threshold, grid refinement is warranted.
3. Grid Refinement via Bisection: Introduce intermediary grid points at the midsections of the quadrilateral's periphery, thereby reducing each grid cell dimension by half.
4. Iterative Refinement to Compliance: Continue the iterative evaluation and bisection until the interpolation error decreases below the threshold or the grid dimensionality conforms to the smallest permissible magnitude.

Figures 2a and 2b illustrate the transition from a coarse grid to a refined grid structure, highlighting the adaptive nature of the method. The procedural sequence ensures that the grid is refined only where necessary, optimizing computational and storage resources. This approach contrasts with traditional methods that apply uniform grid refinement, often leading to unnecessary computational overhead.

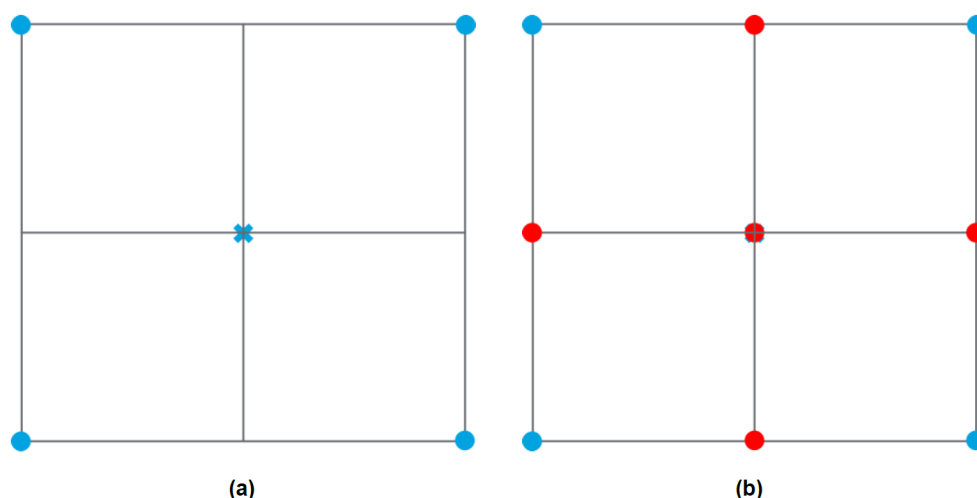


Figure 2—Schematic diagram of the variable grid generation procedure. (a) Initialization with a coarse grid and an assessment of interpolation error at the central intersection (marked in blue cross), utilizing the four adjacent nodal points (marked in red circles); (b) Conditional upon exceeding the predefined error threshold, the grid is refined with five additional points (marked in red circles).

Results

The efficacy of the proposed methodology was validated through a complex synthetic model. Initial tests involved overlaying a coarse grid on a traveltimes map, followed by iterative refinement based on interpolation error thresholds. The method demonstrated adaptive grid adjustments in local regions where interpolation errors exceeded the threshold, thereby safeguarding the precision of traveltimes interpolation and the quality of the resultant seismic image.

Synthetic Model Validation

The method was validated using the SEAM Arid model, a synthetic model designed to simulate realistic geological conditions with varying velocity layers. The SEAM (SEG Advanced Modeling) Arid model is a comprehensive and complex subsurface model developed by the Society of Exploration Geophysicists (SEG) to provide a benchmark for testing and validating seismic imaging techniques [10]. It includes detailed representations of geological features such as faults, stratigraphic layers, and varying rock properties, making it an ideal testbed for evaluating the performance of advanced seismic imaging methods.

The SEAM Arid model is characterized by its intricate subsurface structure, which includes multiple velocity layers that simulate the heterogeneity found in real-world geological settings. These layers vary in seismic velocity, reflecting different rock types and fluid contents. The model also incorporates complex topography, including steeply dipping reflectors and fault zones, which pose significant challenges for traditional seismic imaging techniques.

Figure 3a shows the initial grid configuration, which consists of coarsely and uniformly distributed grid points. This initial setup is typical of traditional methods that use a uniform grid to calculate traveltimes data. However, as the SEAM Arid model includes regions with significant subsurface heterogeneity, the uniform grid approach can lead to interpolation errors and reduced accuracy in traveltimes calculations.

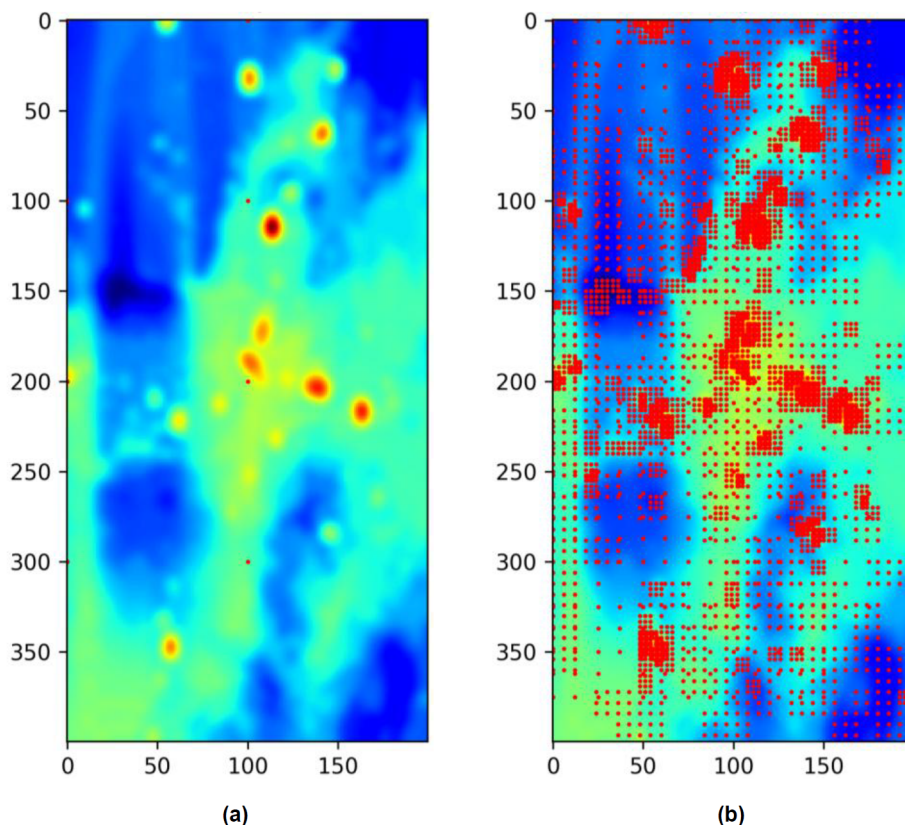


Figure 3—Evaluation of the grid refinement method. (a) Original coarse grid configuration; (b) The grid after refinement, showcasing the increased density where necessary.

To address these issues, the proposed variable grid method was applied to refine the initial grid configuration. The method dynamically adjusts grid density based on interpolation error metrics, ensuring that regions with significant interpolation errors receive higher grid density. This adaptive approach allows for more precise traveltimes calculations in complex geological settings. With an error threshold of 5 milliseconds, the method effectively adjusted grid density in regions with significant interpolation errors, as shown in Figure 3b. The refined grid configuration demonstrates the adaptive nature of the proposed method, with increased grid density in areas where interpolation errors were initially high. This refinement process ensures that traveltimes calculations are accurate and reliable across the entire model.

The adaptive approach of the variable grid method resulted in high precision in traveltimes calculations, leading to a high-quality seismic image. The comparison between the variable grid and a uniform grid achieving the same level of interpolation accuracy revealed a twentyfold reduction in the number of grid points. This significant reduction in grid points translates to substantial savings in computational and storage costs, making the variable grid method a more efficient and cost-effective solution for seismic imaging.

Performance Metrics

The performance of the proposed method was evaluated using several metrics, including computational time, storage requirements, and interpolation accuracy. The results indicated that the variable grid approach significantly outperformed traditional methods in all metrics. Computational time was reduced by approximately 80%, while storage requirements were decreased by a factor of 20. Interpolation accuracy was maintained at a high level, with errors consistently below the predefined threshold.

The reduction in computational and storage demands is particularly beneficial for large-scale seismic imaging projects, where traditional methods can be prohibitively expensive. The proposed method offers a

scalable solution that can be applied to various geological settings, enhancing the efficiency and accuracy of traveltimes calculations.

Discussions

Advantages of the Variable Grid Method

The variable grid method offers several advantages over traditional uniform grid methods. By dynamically adjusting grid density based on interpolation error metrics, the method ensures that computational and storage resources are used efficiently. This is particularly important in areas with complex subsurface structures, where traditional methods may require excessively fine grids to achieve the same level of accuracy.

The adaptive nature of the variable grid method also allows for more precise control over interpolation errors. By refining the grid only in regions where errors exceed a predefined threshold, the method maintains high accuracy while minimizing unnecessary computational overhead. This results in significant reductions in both computational time and storage requirements, making the method well-suited for large-scale seismic imaging projects.

Potential Applications

The proposed variable grid method has potential applications in various fields of geophysics and seismic imaging. In addition to oil and gas exploration, the method can be used in reservoir characterization, geotechnical investigations, and environmental monitoring. The ability to maintain high accuracy while reducing computational and storage demands makes the method particularly valuable in scenarios where large volumes of data need to be processed efficiently.

The method's adaptability to different geological settings also enhances its applicability. By dynamically adjusting grid density based on local interpolation errors, the method can be tailored to the specific requirements of different projects. This flexibility makes the variable grid method a versatile tool for a wide range of geophysical applications.

Limitations and Future Work

While the variable grid method offers significant advantages, there are also some limitations that need to be addressed. One potential limitation is the reliance on interpolation error metrics to guide grid refinement. In areas with highly complex subsurface structures, the interpolation errors may not fully capture the intricacies of the velocity model, potentially leading to incomplete error evaluations.

Future work will focus on further refining the method to address these limitations. Potential improvements include the integration of advanced interpolation techniques and the development of automated grid refinement algorithms. Additionally, the method's adaptability to real-time data acquisition and processing will be investigated, enhancing its applicability to field operations.

Conclusions

The variable grid generation method presented in this study offers a robust solution to the computational and storage challenges associated with traveltimes calculations in seismic imaging. By dynamically adjusting grid density based on interpolation error metrics, the method ensures high precision while significantly reducing computational and storage demands. Numerical tests confirm the effectiveness of this approach in optimizing grid configurations and improving the overall performance of seismic imaging processes.

Future work will focus on further refining the method and exploring its applications in more complex geological scenarios. Potential improvements include the integration of advanced interpolation techniques and the development of automated grid refinement algorithms. Additionally, the method's adaptability to real-time data acquisition and processing will be investigated, enhancing its applicability to field operations.

The proposed method represents a significant advancement in seismic imaging technology, offering a practical and efficient solution to the challenges of traveltime calculations. Its ability to maintain high accuracy while reducing computational and storage costs makes it a valuable tool for geophysical applications, including oil and gas exploration, reservoir characterization, and geotechnical investigations.

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